

Chapter 1: Introduction to Data Science

Data Science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge from structured and unstructured data. It combines statistics, mathematics, programming, and domain expertise. The data science workflow includes data collection, cleaning, exploration, modeling, and communication. Python and R are the most popular languages used by data scientists worldwide.

Chapter 2: Data Wrangling with Pandas

Pandas is a powerful Python library for data manipulation and analysis. It provides DataFrame and Series data structures. Key operations include filtering, grouping, merging, and reshaping data. Handling missing values using fillna and dropna is essential. Pandas integrates well with NumPy for numerical operations and Matplotlib for visualization. Reading data from CSV, Excel, and SQL databases is straightforward with Pandas.

Chapter 3: Data Visualization

Data visualization helps communicate insights from data effectively. Matplotlib is the foundational plotting library in Python. Seaborn provides a high-level interface for drawing attractive statistical graphics. Plotly enables interactive visualizations. Common chart types include bar charts, line plots, scatter plots, histograms, and heatmaps. Choosing the right visualization depends on the type and relationship of the data being analyzed.

Chapter 4: Statistical Analysis

Statistics is the backbone of data science. Descriptive statistics summarize data using mean, median, mode, variance, and standard deviation. Inferential statistics allow conclusions to be drawn from samples. Hypothesis testing, p-values, and confidence intervals are key concepts. Correlation measures the relationship between variables. Probability distributions like normal, binomial, and Poisson are widely used in statistical modeling.

Chapter 5: Machine Learning with Scikit-Learn

Scikit-Learn is a Python library that provides simple tools for machine learning. It includes algorithms for classification, regression, clustering, and dimensionality reduction. The typical workflow involves splitting data, fitting a model, and evaluating performance. Metrics like accuracy, precision, recall, and F1-score measure model quality. Cross-validation and hyperparameter tuning with GridSearchCV improve generalization of models.